MFE FORMWORK MR11 SYSTEM

# Software Requirements Specification

Version 1.0

Prepared for

**MFE Formwork**

Prepared by

**MyIntellX project team**

Date

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Status

**Draft for client review**

# Document control

| Item | Detail |
|---|---|
| Document | Software Requirements Specification |
| System | MFE Formwork MR11 System, version 1.0 |
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| Prepared for | MFE Formwork |
| Prepared by | MyIntellX project team |
| Intended readers | Client project owner, UAT team |
| Classification | Confidential — client and vendor use only |

## Revision history

| Version | Date | Description | Author |
|---|---|---|---|
| 1.0 | 21 October 2026 | First issue for client review | MyIntellX project team |
| | | | |

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# 1. Summary

Version 1.0 meets 47 functional requirements across six modules, plus the non-functional requirements in section 5. Each requirement was checked against the source code at commit `0cc8c36`.

# 2. Purpose and scope

The system replaces the manual, workbook-by-workbook MR11 Master Schedule. Seven departments upload their usual Excel workbooks; the system merges them into one project-by-project schedule, a CEO dashboard and an administration console.

| In scope (version 1.0) | Out of scope |
|---|---|
| Sign-in, password change, accounts managed by administrators | Self-registration, single sign-on |
| Workbook upload and viewing for BD, Finance, Shell Plan & Design, Planning, Production, Dispatch (Local and Overseas) | Editing department data in the browser |
| MR11 Master Schedule with filters, export, and admin row delete and restore | Email or push notifications |
| Series, approval and monthly dispatch history | Mobile app |
| CEO dashboard | ERP or accounting integration |
| Admin console | |

# 3. Stakeholders and user roles

| Role | Uses the system to |
|---|---|
| CEO / executives | Read the CEO dashboard and the MR11 |
| Business Development (BD) | Upload the master project list (contract, area, price, delivery) |
| Finance | Upload payment status and amounts received |
| Shell Plan and Design | Upload consultant drawing status, approvals and design completion |
| Planning | Upload processing series and cumulative processed quantity |
| Production | Upload fabrication output |
| Dispatch | Upload Local and Overseas shipment and delivery data |
| System administrator | Manage users and roles; delete or restore MR11 rows |

# 4. Permissions

| Capability | ADMIN | CEO | Department roles |
|---|---|---|---|
| Sign in, change own password | Yes | Yes | Yes |
| View MR11, export to Excel, sync MR11 | Yes | Yes | Yes |
| Delete or restore MR11 rows | Yes | No | No |

| Capability | ADMIN | CEO | Department roles |
|---|---|---|---|
| View, upload and download a department workbook | All departments | No | Own department only (SHELLPLAN and DESIGN share one workbook) |
| CEO dashboard | Yes | Yes | No |
| Admin console | Yes | No | No |

# 5. Functional requirements

## 5.1 Authentication and accounts (AUTH)

| ID | Requirement |
|---|---|
| AUTH-01 | A user signs in with email and password. The email is matched without regard to case or surrounding spaces. |
| AUTH-02 | A wrong email shows "No account found for this email"; a wrong password shows "Wrong password". |
| AUTH-03 | A successful sign-in issues a session token valid for 7 days. |
| AUTH-04 | A user still on the default password is prompted to change it. |
| AUTH-05 | A user can change their password. The new password must be at least 8 characters, differ from the current one, and not be the default password. |
| AUTH-06 | An expired or invalid session signs the user out and returns them to the sign-in page. |
| AUTH-07 | Each page opens only for users allowed to see it; others are sent to the MR11 page. The server enforces the same rules on its data. |

## 5.2 Department workbooks (DEPT)

| ID | Requirement |
|---|---|
| DEPT-01 | Each department page shows its active workbook as a spreadsheet, keeping values, fill colours, font colours and merged cells. |
| DEPT-02 | An authorised user uploads an .xlsx file of up to 25 MB; it becomes the department's active version. |
| DEPT-03 | Shell Plan and Design share one combined workbook ("Shell Plan & Design"). |
| DEPT-04 | Dispatch keeps two files, Local and Overseas. An upload replaces only its own file. Without a stated part, the system decides by file name, then by columns (ETD / ATD columns mean Overseas). |
| DEPT-05 | Every upload is saved to the database before it becomes active. If saving fails, the uploader sees an error and the previous version stays active. |
| DEPT-06 | Every upload is logged with user, department and file name, and triggers a rebuild of the MR11. |
| DEPT-07 | Day headers stored as Excel date numbers display as dates. |
| DEPT-08 | Planning and Production pages show series history; the Shell Plan & Design page shows approval history; the Dispatch page shows m² dispatched per month. |

## 5.3 MR11 Master Schedule (MR11)

| ID | Requirement |
|---|---|
| MR11-01 | BD is the master list: every MR11 row starts from a BD row with a value in at least one cell. |
| MR11-02 | Rows from other departments are matched to a BD row by project (short name first, then project number), stream, and the closest font and fill colour. |
| MR11-03 | A project number may carry the stream as its last digit (251242 = project 25124, stream 2). |
| MR11-04 | Finance fills Advance Received / Payment Status, Actual Received and Payment Date. |
| MR11-05 | Shell Plan fills its status and approved date; Design fills Formwork Design Status, completion date per level and Total Quantity Ordered m². |
| MR11-06 | Design completion dates are coloured: dark green = completed and passed, yellow = completed but still ahead, red = not completed. |
| MR11-07 | Total Processed is the sum of Planning's Cumulative Processed (Project) for the row's colour blocks. Processed Date is the upload date on which that total last changed. |
| MR11-08 | Total Produced is the sum of Production's Cumulative Produced (Project). Produced Date is the latest daily column with output. |
| MR11-09 | Total Dispatch, Dispatch Date, Formwork Quantity Sailed (m²) and ETD/ATD come from the Dispatch files. Local projects show "Delivered:" dates instead of sailing dates. |
| MR11-10 | A project is Local when it is in Peninsular Malaysia; Sabah, Sarawak, Labuan and other countries are Overseas, unless a Dispatch file already lists it. |
| MR11-11 | Blue-font BD rows (Re-Design Only) stop at Planning; their Production and Dispatch columns show "-". |
| MR11-12 | Monthly columns Jan-26 to Dec-27 show actual m² dispatched for ended months and the forecast for the current and later months, with yearly totals. |
| MR11-13 | Rows of the same project and stream with the same fill colour share merged Shell Plan and Design cells. |
| MR11-14 | Rows sort by project, stream, unfilled before filled, fill colour, then font colour (black first). |
| MR11-15 | Users can search, filter rows, show or hide column groups, zoom and use full screen; the page remembers these settings per browser. |
| MR11-16 | A department user's own columns are highlighted in the department colour. |
| MR11-17 | Export produces an .xlsx file with the two-row header, row colours, number formats and merged cells. |
| MR11-18 | An administrator can delete a row from the MR11 and restore it later; a deleted row stays out of every rebuild until restored. |
| MR11-19 | Any signed-in user can sync, which rebuilds the MR11 from every department's latest workbook. |

## 5.4 Pricing and market data (PRICE)

| ID | Requirement |
|---|---|
| PRICE-01 | Selling Price (USD) combines BD's USD price, MYR price and any other-currency price, converted at today's rate. |
| PRICE-02 | LME Adjusted (USD) = (today's LME aluminium cash price − the row's LME Rate) ÷ 1000 × 21.5 for "Variable" rows; 0 for "Fixed"; "Check" for "Freeze". |
| PRICE-03 | Final Selling Price (USD) = Selling Price + Props, WPB, Waler, Acc + Aluminium Weight Adjusted + Freight Adjusted + LME Adjusted. |

| ID | Requirement |
|---|---|
| PRICE-04 | Prices are recalculated each time the MR11 is read or exported, not only on upload. |
| PRICE-05 | Exchange rates come from Bank Negara Malaysia, with open.er-api.com and the European Central Bank (Frankfurter) as fallbacks; LME prices come from Westmetall. Both are cached for one hour, and the source is shown on screen. |
| PRICE-06 | Other-currency names are recognised by ISO code, English name or symbol (for example SGD, "Singapore Dollar", S$). |

### 5.5 CEO dashboard (CEO)

| ID | Requirement |
|---|---|
| CEO-01 | The dashboard shows stage totals and per-row figures from the latest MR11, filtered by region and month. |
| CEO-02 | A month pipeline shows, per month, the BD forecast, Design, Processed, Produced and Dispatched m². Blue-font rows are labelled Re-Design Only. |
| CEO-03 | A pricing panel shows average selling price per m² with today's LME price and exchange rate. |

### 5.6 Administration (ADMIN)

| ID | Requirement |
|---|---|
| ADMIN-01 | An administrator lists all users with their roles. |
| ADMIN-02 | An administrator creates a user with email, name, password and a starting profile. |
| ADMIN-03 | An administrator changes a user's roles, by profile or role by role. |
| ADMIN-04 | An administrator deletes a user. The root account admin@mfeformwork.com cannot be deleted or lose its ADMIN role. |

## 6. Non-functional requirements

| Area | Requirement |
|---|---|
| Availability | Runs as serverless functions on Vercel (Singapore) or a container on Azure App Service. Health check at `/health`; database check at `/api/health/db`. |
| Data durability | Supabase PostgreSQL is the shared source of truth. Without a database the system runs from memory and loses data on restart (demo mode only). |
| Consistency | Each server instance reloads users, workbooks and the latest MR11 from the database before serving. |
| Security | Passwords stored hashed; every API except sign-in and health checks requires a signed token; role checks run on the server. |
| Performance | A BD upload with rebuild completed in 0.35 s in the pre-UAT test (sample data, local server). |
| Usability | Comma thousands separators; 3 decimal places for decimal numbers; payment amounts in dollars. |
| Time zone | "Today" and month boundaries follow Malaysia time (Asia/Kuala_Lumpur). |

| Area | Requirement |
|---|---|
| Browser support | Current desktop Chrome, Edge, Firefox and Safari. |
| File limits | Upload up to 25 MB per workbook; request body up to 100 MB. |

## 7. Assumptions and constraints

- Departments keep their workbook layouts and header names. A renamed header may stop a column from mapping; the mapping list is in `server/config/mr11.config.ts`.
- Colour conventions (font and fill) carry meaning and stay consistent between departments; small shade differences are tolerated.
- Month columns cover January 2026 to December 2027; later years need a configuration change.
- Market data depends on public services (Bank Negara Malaysia, Westmetall) staying available.

## 8. Glossary

| Term | Meaning |
|---|---|
| MR11 | The master schedule that tracks every project from contract to dispatch |
| Stream | A sub-package of a project, taken from the project number or Stream column |
| Series | A batch of formwork processed (Planning) or fabricated (Production) |
| Font / fill colour | Row text and background colours; they mark blocks, plants and Re-Design Only rows |
| LME | London Metal Exchange aluminium price |
| ETD / ATD | Estimated / actual time of departure |
| RBAC | Role-based access control |